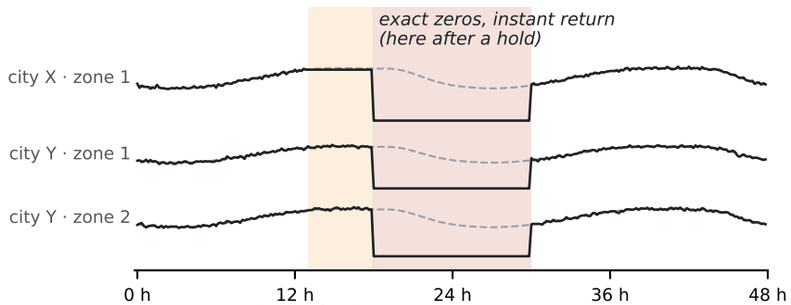
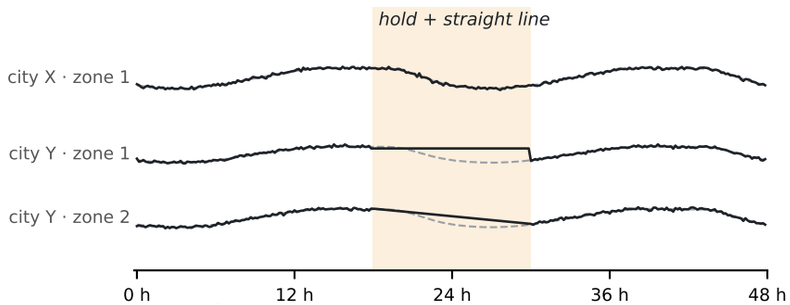
Fig 2 — Typology: four artefacts and five grid events, each with a measurable signature at a level of the hierarchy (schematic; straight-line fills are part of A2 and A3)

A1 · zero-fill gap



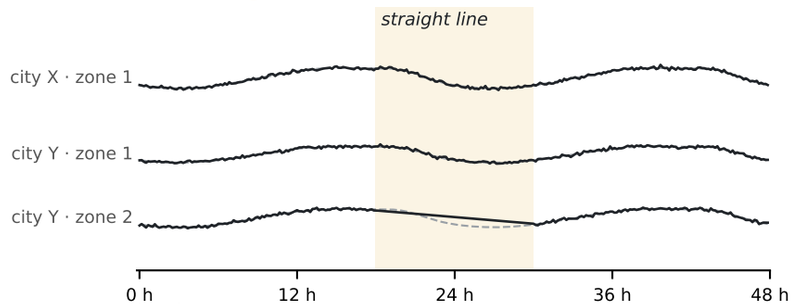
level: region (≥ 2 cities) or episode
signature: exact zeros (η = 1) in ≥ 2 cities at once, often after a hold or straight-line fill; returns at once
rules: R1, R3, R4, R21

A2 · simultaneous fill



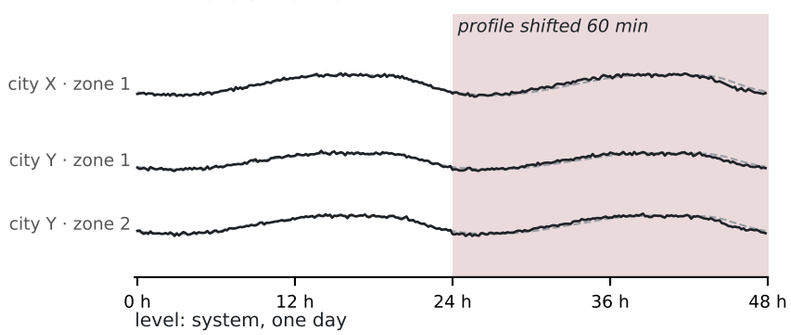
level: ≥ 2 zones
signature: ≥ 2 zones frozen (identical values) or straight-line interpolated over the same span
rules: R2, R19

A3 · single-zone fill



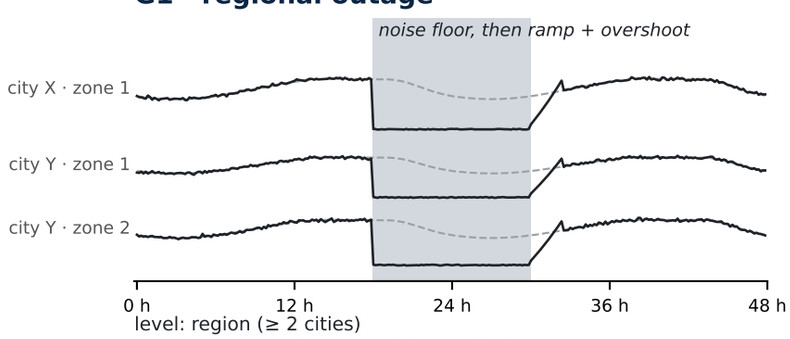
level: zone
signature: one zone frozen or straight-line interpolated; neighbours move normally
rules: R12, R20

A4 · clock error



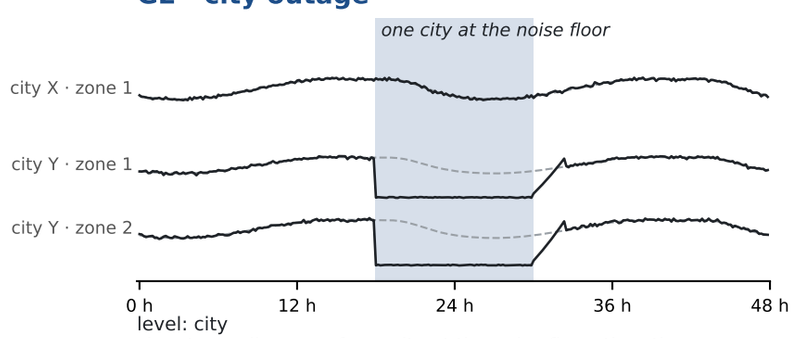
level: system, one day
signature: daily profile shifted by δ* ≈ ±60 min after a clock-change date; duplicated or missing stamps
rules: R13, R14, R15

G1 · regional outage



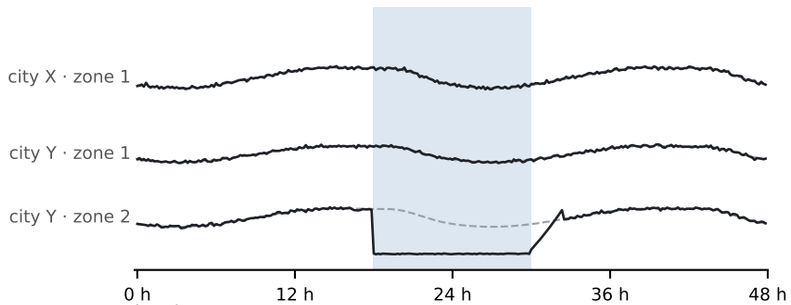
level: region (≥ 2 cities)
signature: noise floor (η < 0.9) in ≥ 2 cities, ramp restoration with cold-load overshoot π ≥ 1.15
rules: R5, R6, R7

G2 · city outage



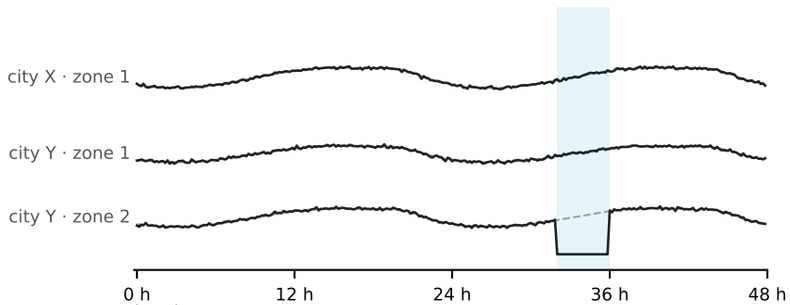
level: city
signature: all zones of one city at the noise floor, the other cities normal
rules: R8 (+ R5, R6)

G3 · unplanned feeder outage



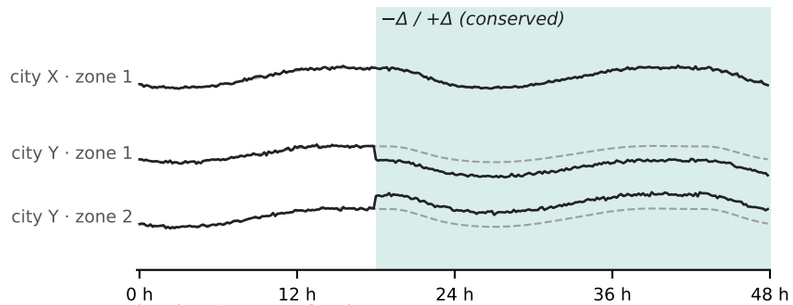
level: zone
signature: one zone at the noise floor at any hour, possibly > 24 h, ramp restoration
rules: R9, R11 (+ R6)

G4 · planned work



level: zone
signature: one zone at the noise floor, starting 06:00–11:59, short (< 6 h)
rules: R10

G5 · load transfer



level: ≥ 2 zones of a city
signature: one zone steps down while another steps up by the same amount (conservation κ ≈ 0)
rules: R16

dashed: expected value x̂ (eq. 1) · black: observed · shaded: the event span · artefacts A1–A4 are data failures (impute); grid events G1–G5 are real (keep and flag)